Empathy Map

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

Primary persona: **the lecturer**, the main user of the dashboard. A second map covers the **student**, the person being observed, because the system's privacy design is built around their concerns.

Persona 1: Lecturer (Primary User)

Teaches 60-120 students per lecture and wants more teaching time and honest feedback on how the class is responding.

SAYS

- "I lose ten minutes on roll call."
- "Someone signed in for their friend again."
- "Are they even following this slide?"

THINKS

- Will I finish the syllabus on time?
- Is the class lost, or just quiet?
- Can I trust a camera system with my students' data?

DOES

- Calls names or passes around a sheet.
- Scans faces for signs of boredom.
- Rushes explanations when time runs short.

FEELS

- Frustrated by wasted time and proxies.
- Anxious about missing struggling students.
- Hopeful that a tool could handle the routine work.

PAINS	GAINS
Slow roll calls and proxy attendance. No objective attention data. Fear of privacy complaints. Tools that need internet or special hardware.	Attendance done automatically in the background. A live engagement view and clear teaching tips. Local-only processing, so privacy is easy to defend. One-click CSV export and manual corrections.

Persona 2: Student (Observed User)

Wants attendance recorded fairly and does not want to be watched by an unknown system.

SAYS

- "Why is there a camera pointed at us?"
- "I was there the whole time, why am I marked absent?"
- "Can I say no to my face being stored?"

THINKS

- Where does my face data go?
- Will looking at my notes count against me?
- Is this fair if my seat is badly lit?

DOES

- Looks down to take notes.
- Arrives a few minutes late.
- Asks classmates what the system does.

FEELS

- Uneasy about surveillance.
- Relieved when privacy is explained clearly.
- Annoyed by unfair marks.

PAINS	GAINS
Biometric data stored without consent. Being penalized for blinks or note-taking. No way to contest a wrong record.	Biometric enrollment only with consent; deregistration purges data. Anonymous tracking (Student #1) for non-enrolled people. Smoothing so brief glances do not hurt scores; lecturer can override.

How the System Responds

Insight from the empathy map	Design response in the system
Lecturer loses time and cannot detect proxies	Presence-based attendance accumulated over the whole session
Lecturer wants to know if the class is attentive	Composite engagement score (gaze 40%, pose 20%, presence 20%, audio 20%)
Lecturer fears alert overload	Teaching recommendations with a 30-second debounce and priority levels
Student fears surveillance	100% offline processing, consent-gated enrollment, no raw video stored
Student fears unfair penalties	EMA smoothing (beta = 0.70), manual override with audit reason